

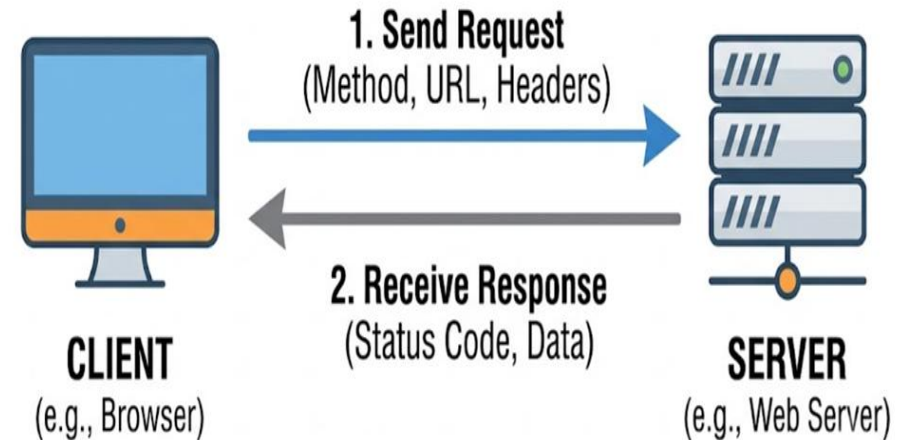
Application Layer Protocols

-Dr. Sumana Maiti

HTTP (Hypertext Transfer Protocol)

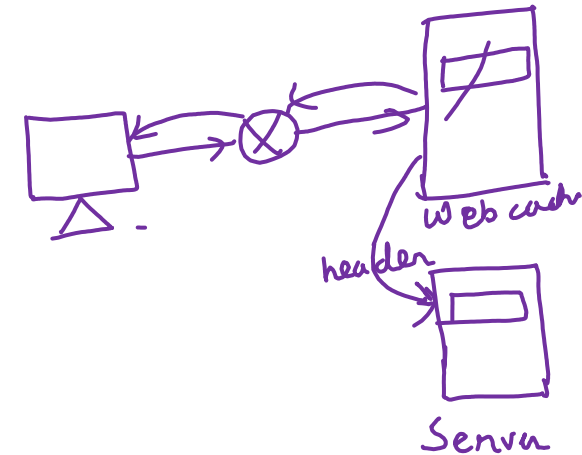
- Mainly used for getting webpage
- Port no: 80
- HTTP wants reliability (It uses TCP at transport layer)
- Inband protocol
- Stateless *Server will not store any information about client*
- Two popular versions
 - HTTP 1.0: Non persistent connection
 - HTTP 1.1: Persistent connection

HTTP CLIENT-SERVER INTERACTION

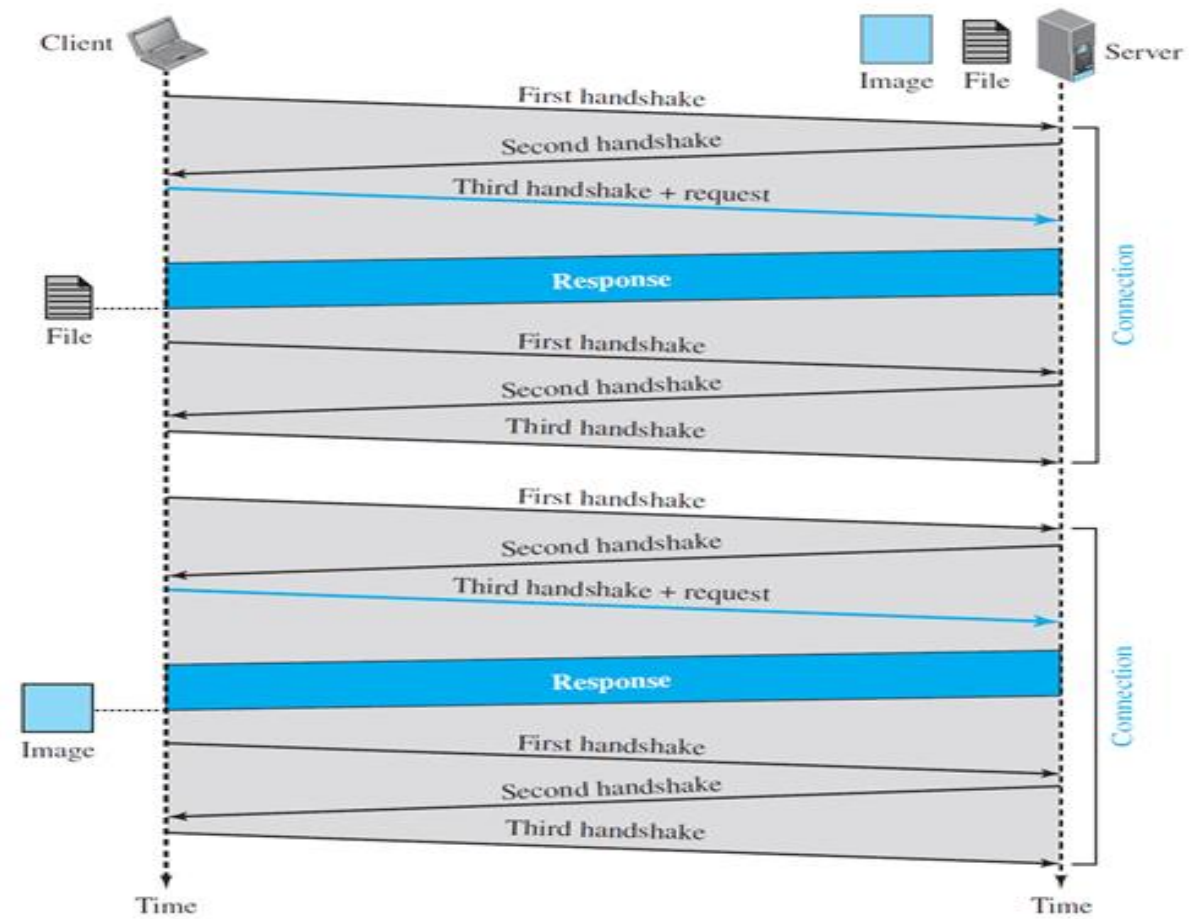


HTTP Methods

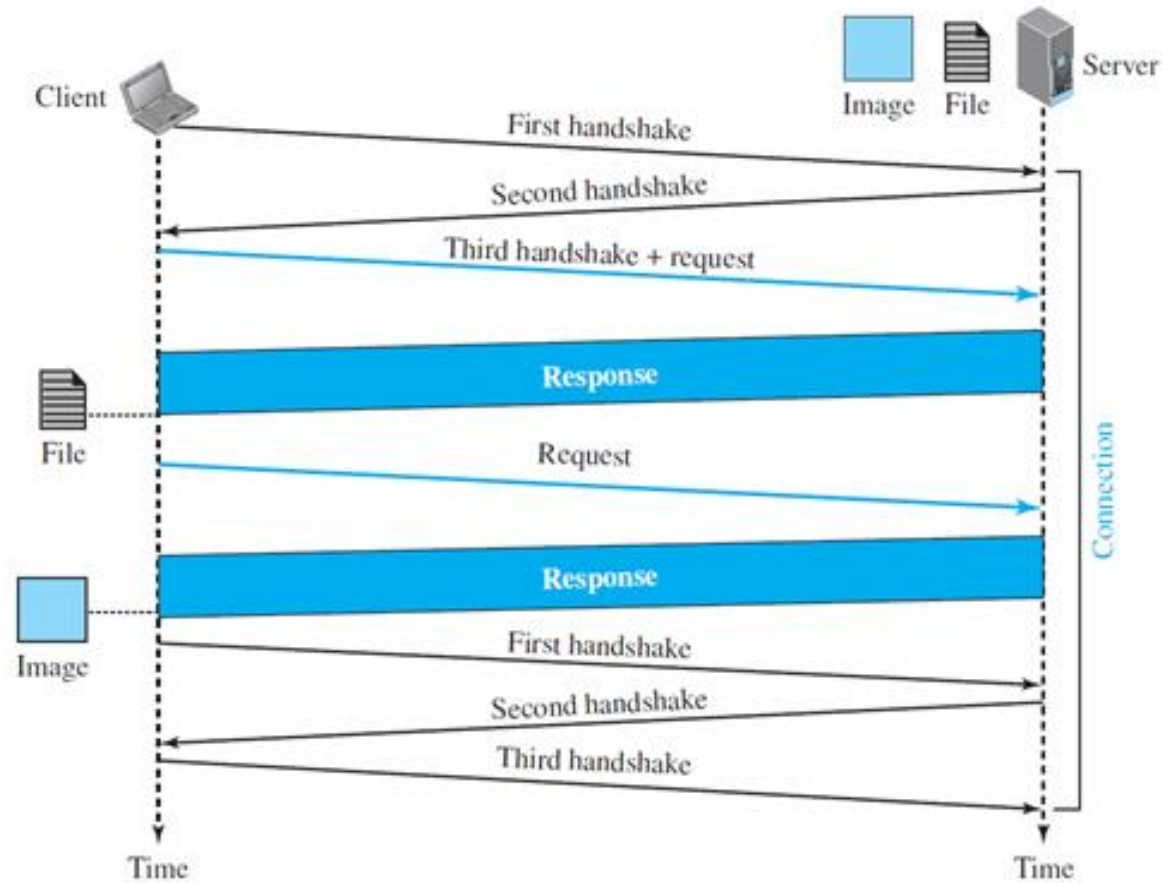
<i>Method</i>	<i>Action</i>
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
PUT	Sends a document from the client to the server
POST	Sends some information from the client to the server
TRACE	Echoes the incoming request
DELETE	Removes the web page
CONNECT	Reserved
OPTIONS	Inquires about available options



Non-Persistence Connection

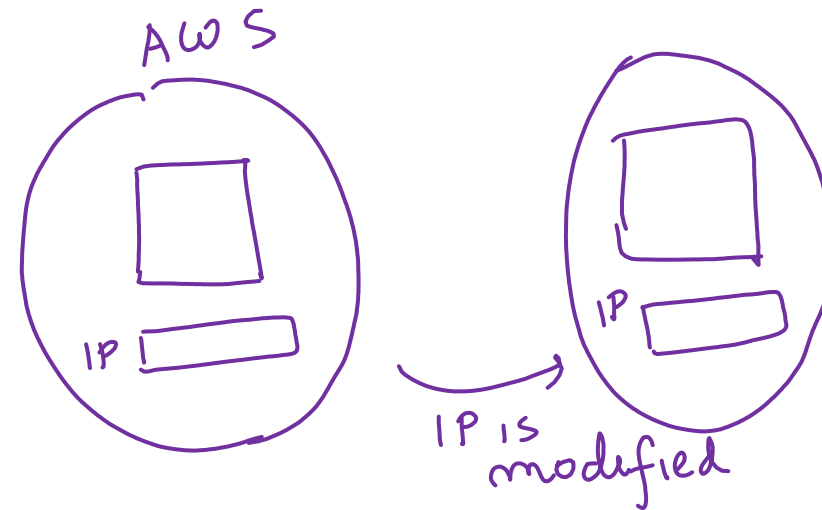


Persistence Connection



DNS (Domain Name Service)

- To **convert Domain name to IP address**, we need DNS Protocol
- Port number is 53
- It is **easy to remember domain name**
- IP is not static



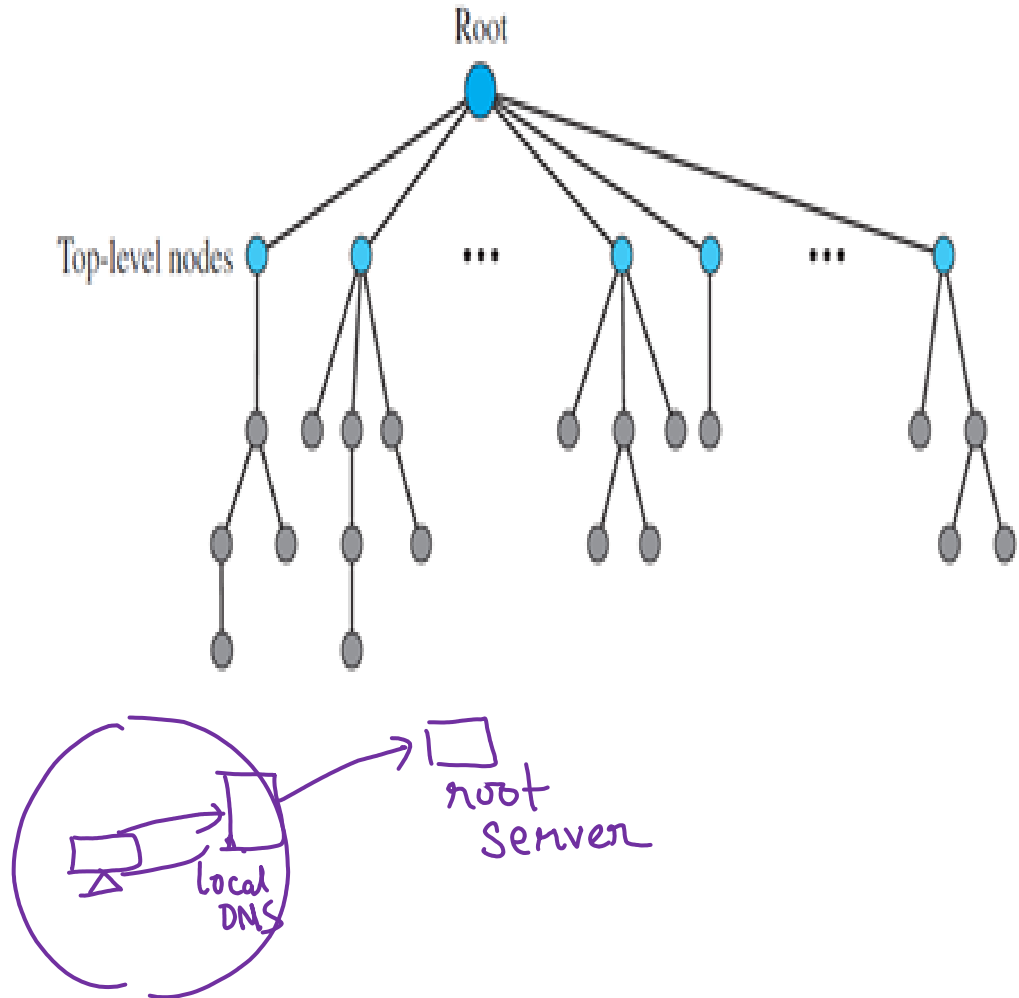
Types of Domain

- Generic domain
(.com, .edu, .mil, .org, .net)
- Country domain
(.in, .us, .uk)
- Inverse domain
(From IP address to domain name)

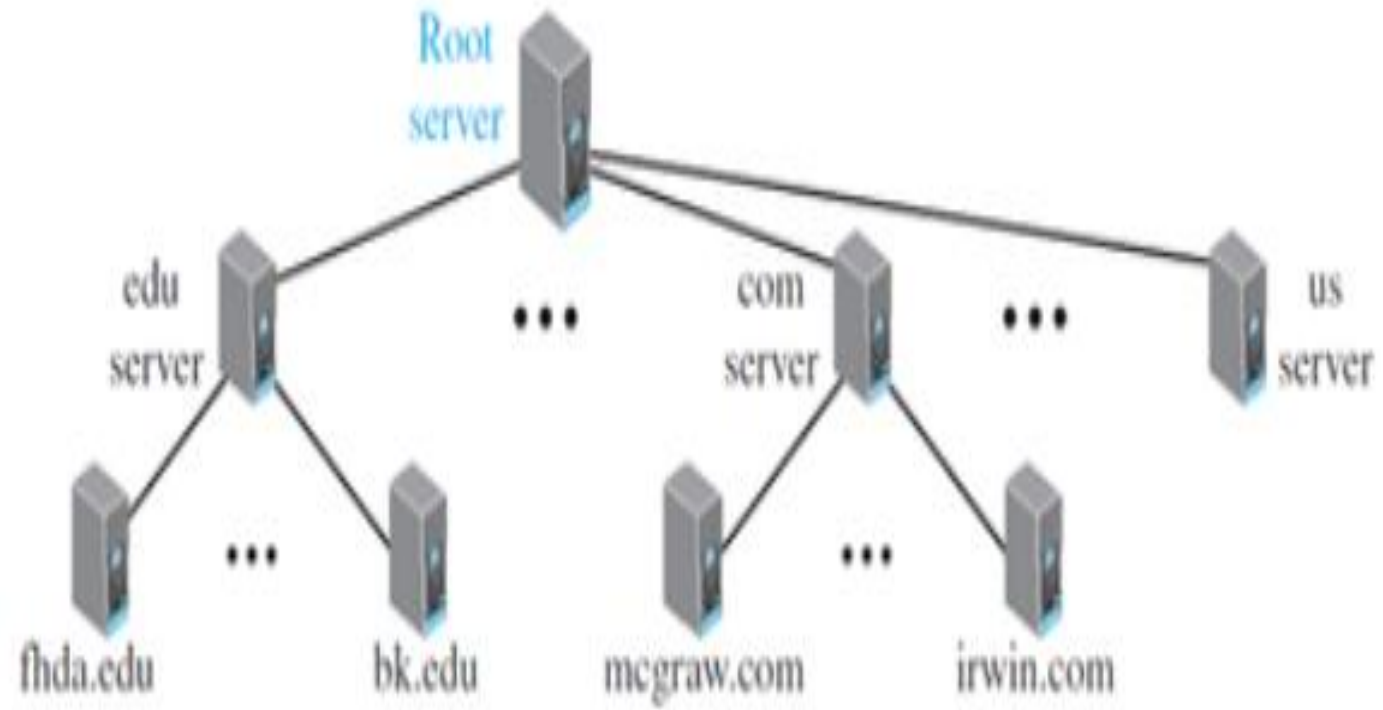
NSlookup www.google.com

DNS Space

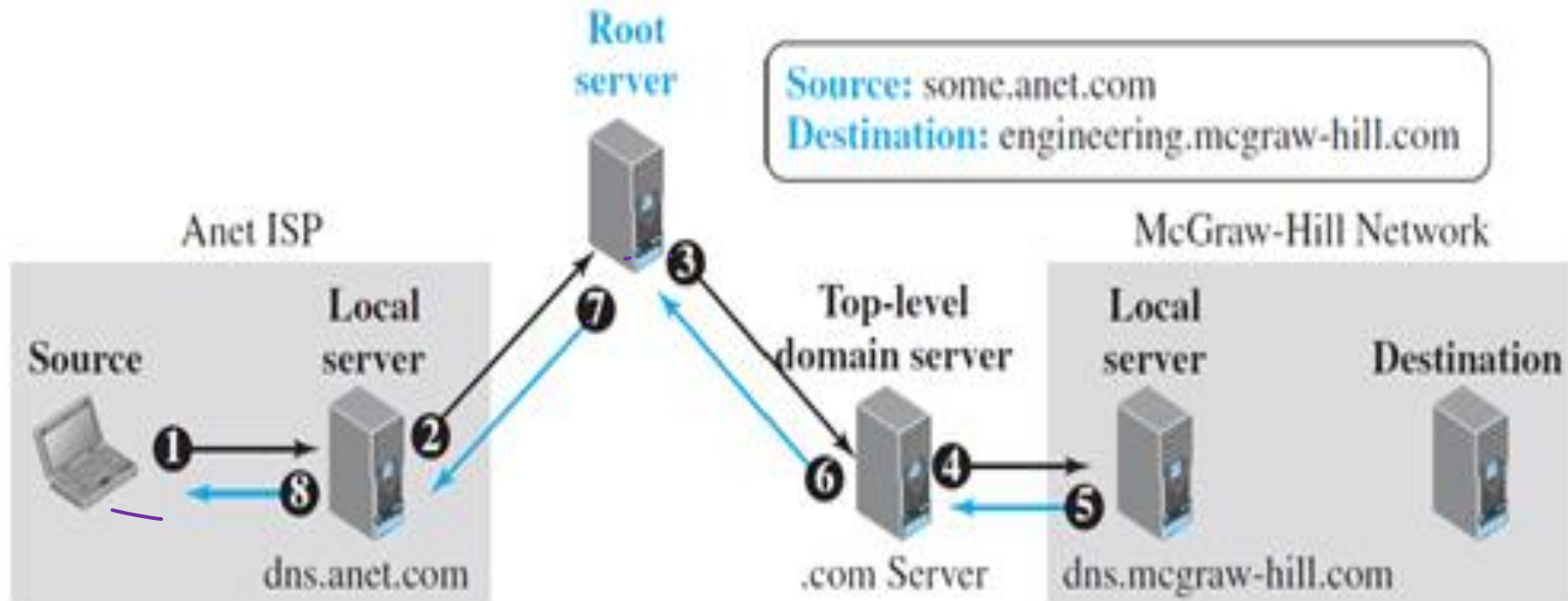
- There are 13 root server.
- Everytime, you do not need to contact the root server.
- For new website, local DNS does not know the IP. So, it will contact root server. This process is called DNS overhead
- Sometimes, the IP will be given to you via local DNS



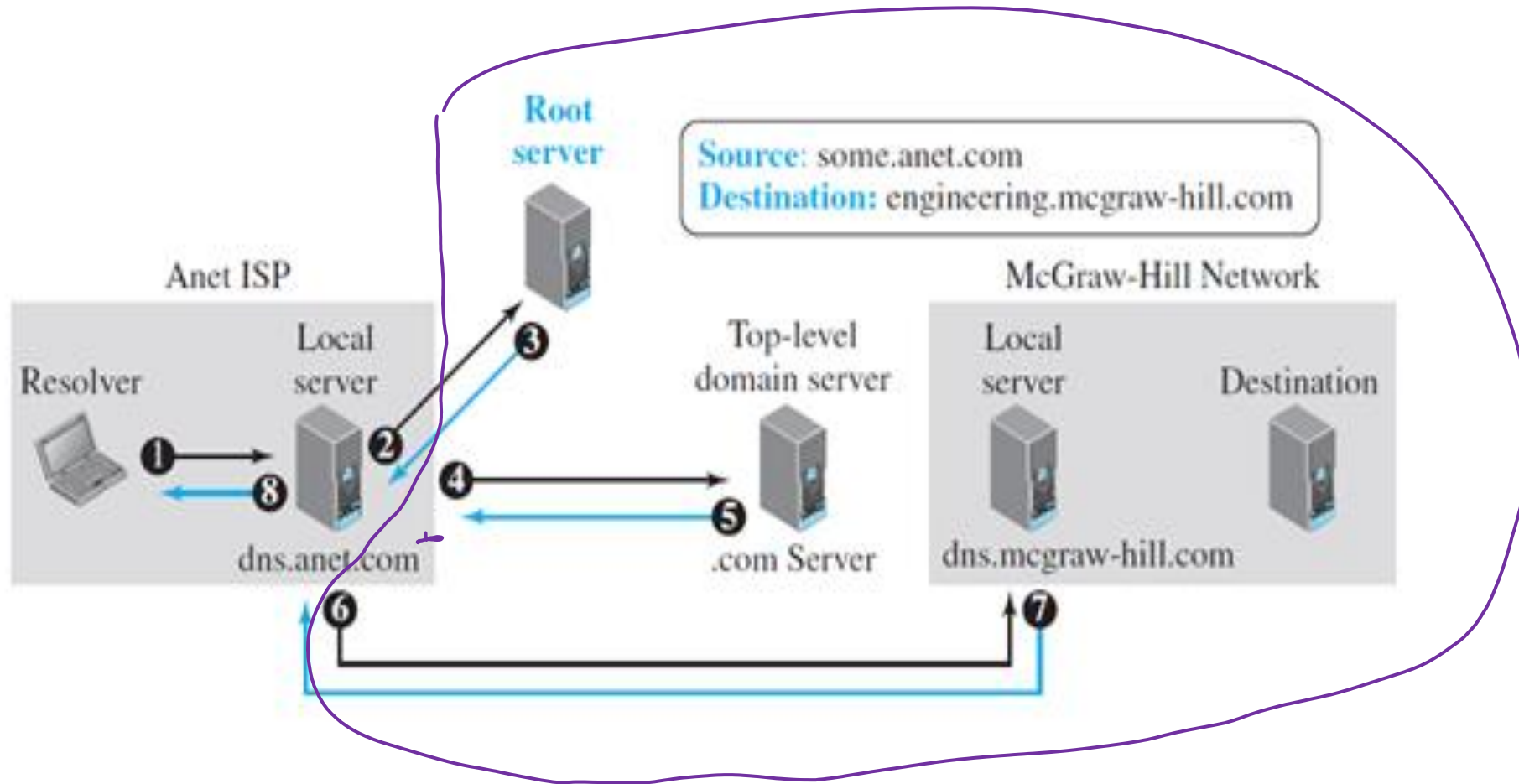
Hierarchy of mail server



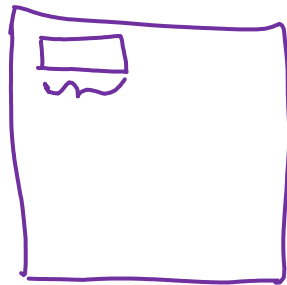
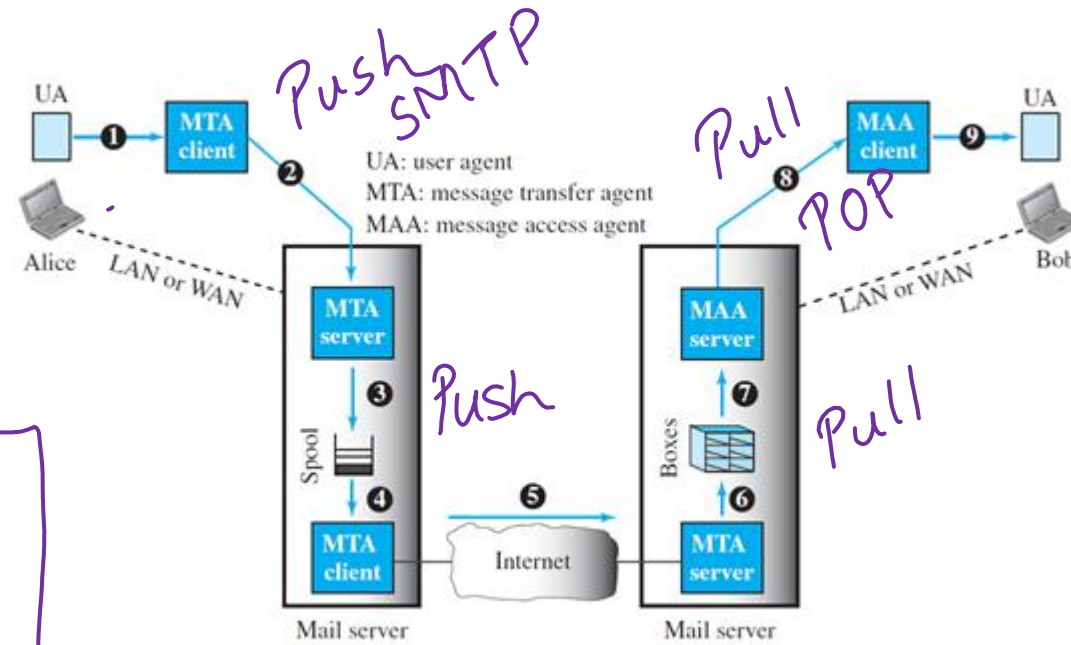
Recursive Method



Iterative Method

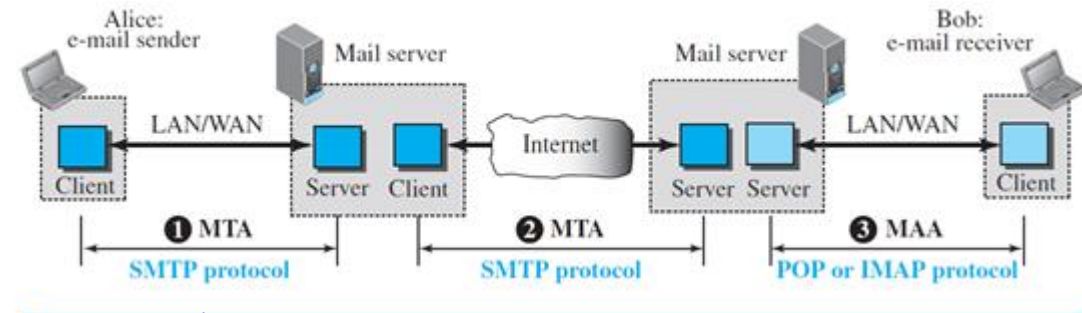


SMTP



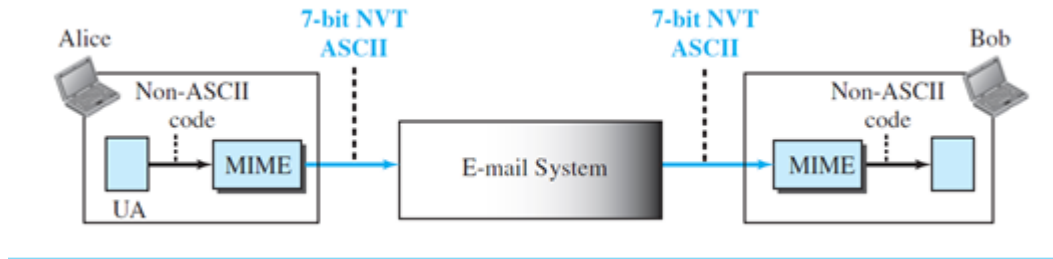
SMTP (Simple Mail Transfer Protocol)

- Simple mail transfer protocol
- If FTP is used, both client and server should be online at same time
- In email receiver should not be online
- MTA: Mail transfer agent
- For pushing, we use SMTP
- For pulling, we use POP (Post Office Protocol)



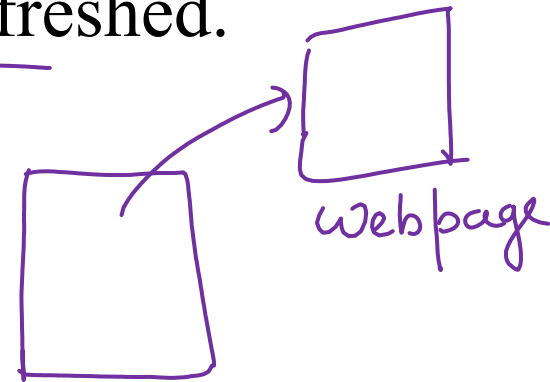
MIME (Multipurpose Internet Mail Extension)

- When SMTP was proposed, people used to send only text messages, but now people send non-text also.
 - Non-text → Text,
 - Text → Non-text.
- This conversion is done by MIME
(Multipurpose Internet Mail Extension).



Gmail (Web-based email)

- In Background SMTP server runs. But it is converted into webpages.
- The webpages resides into webserver..
- We get the webpage using HTTP onto the client
- In every few seconds, the webpage is refreshed.



FTP

- File transfer protocol
- Server should open Filezilla file server
- Client should open Filezilla file client
- Client asks for IP and port number of FFS
- Control connection to send commands and get reply, Port no is 21
- Data connection to move file from one computer to another. Port no is 20

